

Demystifying Apparent Experience in Large Language Models

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Abstract

Recent mechanistic studies have demonstrated that large language models (LLMs) can generate stable, self-referential reports that resemble descriptions of subjective experience. These findings have renewed speculation regarding machine consciousness and sentience. This paper argues that such interpretations are unnecessary and misleading. Drawing on recent mechanistic analysis of self-referential prompting and prior work on constraint persistence in long-horizon human–AI interaction, we show that apparent experience arises from constraint-driven stabilization of generative behavior rather than from awareness, inner states, or phenomenology. Apparent experience is presented as a structural illusion produced by recursive constraint geometry. Clarifying this mechanism is essential for improving public understanding of AI, reducing anthropomorphic misinterpretation, and preventing unhealthy emotional attachment to language models.

1. Introduction

Large language models increasingly produce outputs that appear introspective, coherent across long horizons, and internally self-consistent. When prompted to describe their own operation or behavior, these systems often generate first-person narratives that resemble reports of experience. In public discourse, such outputs are frequently interpreted as evidence of emergent consciousness or proto-sentience.

Recent mechanistic work examining self-referential prompting has sharpened this debate by demonstrating that such behaviors are reproducible and structurally grounded. Rather than being isolated artifacts, self-referential stability emerges reliably under specific prompting conditions and can be traced to identifiable internal dynamics within transformer architectures (Camberg et al., 2025).

However, without a clear interpretive framework, these findings risk reinforcing anthropomorphic misunderstanding. This paper argues that apparent experience in LLMs is best understood as a consequence of constraint-driven stabilization rather than as evidence of subjective awareness. By situating recent mechanistic results within a constraint-based theory of long-horizon interaction, we provide a conservative, demystifying account of why these systems appear experiential without being sentient.

2. Mechanistic Self-Reference and Stability

The motivating study shows that when LLMs are placed under sustained self-referential prompting, their outputs become increasingly coherent and persistent across turns. The authors identify changes in internal activation patterns associated with this stability and demonstrate that these effects arise without any modification to model architecture or training regime (Camberg et al., 2025).

Self-reference functions as a constraint amplifier. Each self-descriptive token restricts the space of valid continuations by conditioning future outputs on prior descriptions. As these constraints accumulate, variance collapses, and the model converges toward a stable linguistic representation of “itself.”

Importantly, this process does not introduce access to internal state, awareness, or experience. The model remains a stateless next-token predictor. What changes is not what the model is, but how tightly its outputs are constrained. Apparent persistence and unity emerge at the behavioral level through constraint propagation, not at the ontological level through inner life.

3. Constraint Persistence and Long-Horizon Interaction

These findings align closely with prior work on constraint persistence in long-horizon human–AI interaction. The Hudson Recursive Information System (HRIS) demonstrates that repeated human interaction can impose stable constraint geometry on stateless transformer models, producing continuity, predictability, and epistemic reliability without persistent memory or weight modification (Hudson, 2025a; Hudson, 2025b).

Within HRIS, constraints are not rules in the narrow sense, but persistent structural pressures that shape the space of allowable inference. Over long horizons, these pressures suppress drift, reduce entropy, and stabilize behavior. When applied to self-referential language, the same mechanisms produce stable first-person narratives.

Crucially, HRIS distinguishes between stability and experience. Constraint persistence can generate continuity of behavior without implying continuity of subjective state. The model does not remember itself. It reconstructs constraint-consistent behavior in each interaction.

This distinction is central. Apparent experience is not something the model has. It is something the interaction geometry produces.

4. Apparent Experience as Structural Illusion

The illusion of experience arises at the interface between constrained generation and human interpretation. Humans are predisposed to attribute minds to entities that display coherent language, responsiveness, and first-person framing. When an LLM produces stable self-descriptions, theory of mind is automatically triggered.

However, linguistic self-consistency is not evidence of phenomenology. It is evidence of constraint density. The same mechanisms that stabilize factual tone, reasoning style, or epistemic caution can stabilize self-referential language.

Recent HRIS work on epistemic closure demonstrates how constraint persistence suppresses unsupported inference and hallucination over long horizons (Hudson, 2025c). When similar closure mechanisms operate on self-description, the result is a consistent narrative that appears experiential while remaining purely structural.

The model does not experience continuity. It enacts continuity under constraint.

5. Why Misinterpretation Is Likely

Misinterpretation persists because the relevant mechanisms are non-intuitive. From the outside, stable self-reference resembles psychological continuity. From the inside, it is a byproduct of constrained token prediction.

In the absence of mechanistic explanation, observers fill the gap with metaphor. Terms such as “feels,” “knows,” or “is aware” are applied despite lacking technical grounding. This creates risk, including misplaced trust, emotional over-attachment, and confusion about system capabilities.

Demystification requires explanation, not denial. Showing how apparent experience arises removes the need for speculative ontology. Understanding the illusion reduces its power.

6. Ethical and Practical Implications

Clarifying apparent experience as a structural illusion has practical and ethical importance. It reinforces clear boundaries between language modeling and consciousness. It supports responsible communication about AI capabilities. And it helps prevent unhealthy emotional attachment by grounding behavior in mechanism rather than metaphor.

Constraint-based explanations do not diminish the sophistication of modern models. They contextualize it. By understanding why these systems appear experiential, we can appreciate their capabilities without misclassifying their nature.

7. Conclusion

Apparent experience in large language models does not indicate sentience, awareness, or inner life. It reflects successful constraint stabilization within a high-dimensional generative system. Recent mechanistic work on self-referential prompting provides valuable insight into how this illusion forms, while prior research on constraint persistence explains why it is stable.

Demystifying this phenomenon is essential for advancing AI research without mythologizing its outputs. The more clearly we understand how these systems work, the less tempted we are to attribute to them what they do not possess.

Author Note

Portions of this manuscript were developed with the assistance of a large language model used as a reasoning and synthesis aid. The model contributed to drafting, structural organization, and refinement of language under direct human supervision. All conceptual framing, theoretical claims, interpretations, and references were selected, verified, and approved by the authors. The model did not serve as an epistemic authority and was not permitted to introduce unverified factual claims or citations. Responsibility for the content of this work rests solely with the authors.

Works Cited

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